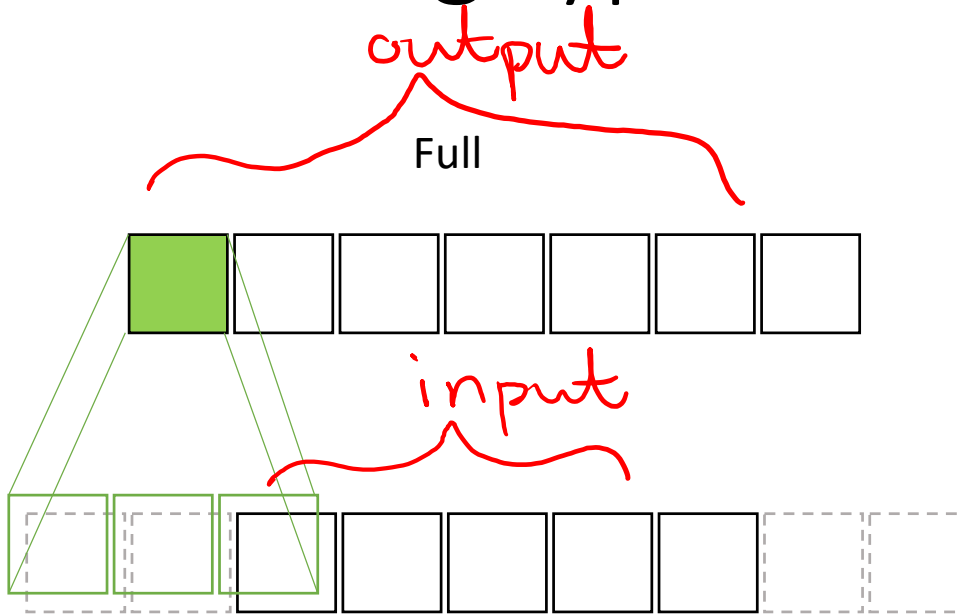


EEE4114F

Padding



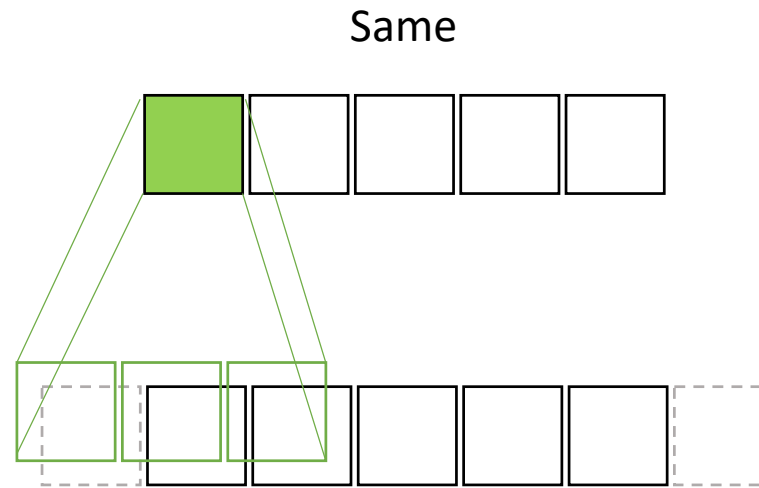
Padding types



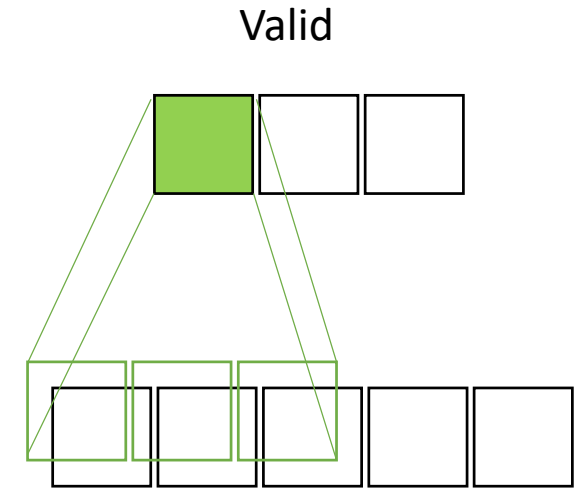
Output is larger than input.
Not often used.

$$P = K - 1$$

kernel size



Input and output size are equal.
Tensorflow can calculate this automatically, but PyTorch can't (manual padding calculation required).



Output is smaller than input.
This may cause problems at the boundaries.
Reducing layer size in this way can be problematic for very deep networks



2D Example

$P=1$ $W=3$

$W=3$

0	0	0	0	0
0	2	1	2	0
0	5	0	1	0
0	1	7	3	0
0	0	0	0	0

X

W $K=3$

0.5	0.7	0.4
0.3	0.4	0.1
0.5	1	0.5

*

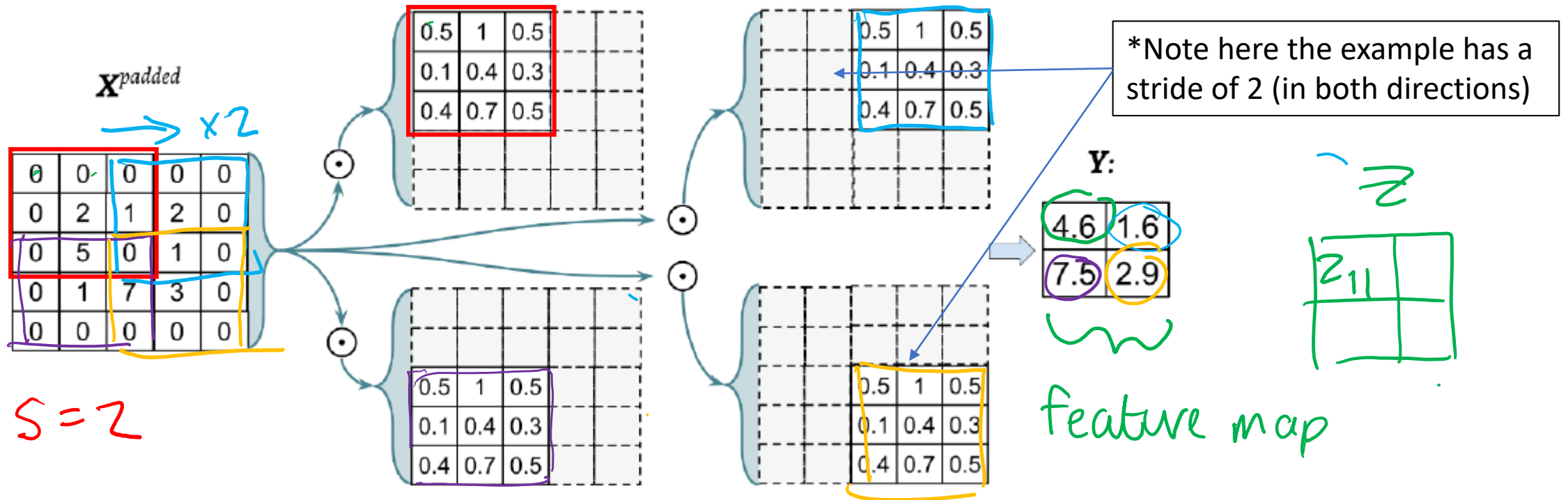
$P=1, K=3, W=3, S=2$

$$\begin{aligned}
 O &= \left\lfloor \frac{W - K + 2P}{S} \right\rfloor + 1 \\
 &= \left\lfloor \frac{3 - 3 + 2(1)}{2} \right\rfloor + 1 \\
 &= 2
 \end{aligned}$$



2D Example

W^r is the rotated kernel (convolution) *not transpose



$S=2$

$$z_{11} = \sum X_{11} \odot W^r$$

$$= 0 \times 0.5 + 0 \times 1 + 0 \times 0.5 + 0 \times 0.1 + 2 \times 0.4 + 1 \times 0.3 + 0 \times 0.4 + 5 \times 0.7 + 0 \times 0.5 = 4.6$$

Next up:

Subsampling

